FACULTY OF ENGINEERING

Your Project Title Here

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Submitted in pursuit of the degree

BACHELOR OF ENGINEERING

In

MECHATRONIC ENGINEERING

North-West University Potchefstroom Campus

Abstract

This report presents the design, development, and testing of ... [Replace this with a concise 200–350 word summary covering: the problem addressed, design approach, key results, and main conclusions.]

Keywords

Keyword One, Keyword Two, Keyword Three, Keyword Four, Keyword Five

Contents

List of Figures

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Declaration of Originality

I, Mr/Ms Firstname Lastname, with student number 12345678 declare the following:

This report is my own original work. I further declare that:

1. No part of it has been copied from another person;
2. I did not work with another person on this project and report;
3. I acknowledged all consulted sources in the text and submitted a bibliography;
4. I have indicated all sections where I have used generative AI-enabled technology;
5. Parts without references are my own ideas, arguments, and conclusions.

The Turnitin plagiarism test resulted in a score of X%.

____

Signature

November 2025

1 Introduction

This report outlines the design, development, and testing of ... [Provide context for the project, explain the motivation, and describe the overall structure of the report.]

The report is structured as follows: Chapter 2 defines the problem and requirements. Chapter 3 presents the literature study. Chapters 4–6 cover the concept, preliminary, and detailed design phases. Chapter 7 describes the implementation. Chapter 8 presents the testing and results. Chapter 9 provides the conclusions.

2 Problem Analysis

2.1 Background

[Describe the domain context and motivation for the project. Explain why the problem exists and why it is important to solve it.]

2.2 Problem Statement

[Clearly state the engineering problem to be solved. Be specific and concise – one or two paragraphs.]

2.3 Project Objectives

The objectives of this project are:

- Objective 1: ...
- Objective 2: ...
- Objective 3: ...

2.4 Requirements

2.4.1 Functional Requirements

- **REQ_01_FUNC**: [Requirement description.]
- **REQ_02_FUNC**: [Requirement description.]
- **REQ_03_FUNC**: [Requirement description.]

2.4.2 Performance Requirements

- **REQ_04_PERF**: [Requirement description.]
- **REQ_05_PERF**: [Requirement description.]

2.4.3 Physical Requirements

- **REQ_06_PHYS**: [Requirement description.]
- **REQ_07_PHYS**: [Requirement description.]

2.4.4 Non-Functional Requirements

- **REQ_08_USABILITY**: [Requirement description.]
- **REQ_09_MODULARITY**: [Requirement description.]

2.5 Risk Analysis

[Summarise the risk analysis methodology and key findings. Reference the detailed risk register in Appendix B.]

3 Literature Study

3.1 Case Studies

[Review relevant published work, prior art, and case studies. Each sub-section can address a different case or research area.]

3.1.1 Case Study 1 – [Topic]

[Discussion with appropriate citations [1].]

3.1.2 Case Study 2 – [Topic]

[Discussion with appropriate citations [2].]

3.2 Existing Solutions

[Compare and critique existing commercial or academic solutions relevant to your project.]

3.3 Possible Solutions

[Identify two or three candidate approaches. Discuss trade-offs and justify the selected approach.]

3.4 Summary

[Brief paragraph drawing together insights from the literature that informed the design decisions.]

4 Concept Design

[Present the high-level functional architecture of the solution. Use functional flow diagrams and operational level diagrams to describe what the system must do without specifying how.]

Functional Flow Diagram

Figure 4.1. *Functional flow diagram illustrating the life-cycle.*

[Table 4.1 lists the functional units identified in the concept design.]

Table 4.1. *Operational Functional Units*

F/U ID	**Name**	**Description**
F/U 1.0	[Name]	[Description]
F/U 2.0	[Name]	[Description]
F/U 3.0	[Name]	[Description]

5 Preliminary Design

[Present the architectural decisions made at the subsystem level. Describe which technologies, components, or strategies were considered for each functional unit and justify the selections.]

5.1 Preliminary Hardware Decisions

5.1.1 [Subsystem 1 – e.g. Processing Unit]

[Discussion of candidate components. Summarise in a comparison table.]

Table 5.1. *[Subsystem 1] Component Comparison*

Component	Criterion A	Criterion B	Selected
Option 1			
Option 2			✓
Option 3			

5.1.2 [Subsystem 2]

[Discussion and selected component with justification [3].]

Figure 5.1. *[Selected component name and description].*

6 Detailed Design

6.1 Detailed Hardware Decisions

[Specify exact components, part numbers, electrical parameters, and interfaces. Include schematics and specification tables.]

Table 6.1. *[Component] Specifications*

Parameter	**Value**
Supply voltage	5 V
Operating current	500 mA
Dimensions	85 mm × 56 mm
Mass	45 g

6.2 Detailed Software Decisions

[Describe software architecture, programming languages, libraries, algorithms, and key design patterns selected.]

6.3 CAD Designs

[Present mechanical CAD drawings for any custom-fabricated parts. Include isometric views and critical dimensions.]

Figure 6.1. *Isometric CAD drawing of [assembly name].*

7 Implementation

7.1 Electrical Implementation

[Describe the electrical assembly process, wiring, PCB fabrication, and any calibration steps. Include wiring diagrams.]

Figure 7.1. *Electrical hardware component layout and wiring diagram.*

7.2 Mechanical Implementation

[Describe the physical construction, materials, machining, and assembly of mechanical components.]

7.3 Software Implementation

[Describe how the software was implemented. Include a high-level flow diagram and, where appropriate, annotated code snippets.]

```
import sensor_library as sl

def init_sensor(pin: int, baud: int = 9600) -> sl.Sensor:
    """Initialise␣and␣calibrate␣the␣sensor␣on␣the␣given␣pin."""
    sensor = sl.Sensor(pin=pin, baud=baud)
    sensor.calibrate()
    return sensor
```

Listing 7.1: Example: sensor initialisation routine.

7.4 Final System

[Present photographs of the completed system and a brief description of its final form.]

(a) *Front view.*

(b) *Side view.*

Figure 7.2. *Final assembled system.*

8 Testing

[Each test should trace back to a specific requirement. State the test procedure, expected outcome, measured result, and pass/fail verdict.]

8.1 Test REQ_01_FUNC – [Requirement Name]

Objective: Verify that . . .

Procedure:

1. Step 1 . . .
2. Step 2 . . .
3. Record measurement.

Results:

Table 8.1. *Results for Test REQ_01_FUNC*

Trial	Expected	Measured	Pass/Fail
1			Pass
2			Pass
3			Pass

Conclusion: The requirement was [met / not met] because . . .

8.2 Test REQ_04_PERF – [Requirement Name]

[Repeat pattern for each requirement under test.]

9 Conclusion

[Summarise what was achieved relative to the objectives stated in Chapter 2. Discuss limitations and suggest avenues for future work.]

References

[1] A. Author, B. Author, "Title of journal article," *Journal Name*, vol. X, no. Y, pp. ZZ–ZZ, Year. doi: `10.xxxx/xxxxxxx`.

[2] C. Author, *Title of Book*. City: Publisher, Year.

[3] D. Author, E. Author, and F. Author, "Title of conference paper," in *Proc. Conference Name*, City, Country, Year, pp. ZZ–ZZ.

[4] Manufacturer Name, *Product Datasheet: [Part Number]*, [Online]. Available: https://www.example.com/datasheet.pdf. [Accessed: Day Month Year].

A Generative AI Usage

Table A.1. *Generative AI (GA) evidence throughout the report*

Section	Tool	Prompt	How output was used

B Risk Register

Table B.1. *Risk Identification and Management*

ID	Risk	Likelihood	Impact	Mitigation
R01	[Description]	Low	High	[Mitigation strategy]
R02	[Description]	Medium	Medium	[Mitigation strategy]
R03	[Description]	High	Low	[Mitigation strategy]

Table B.2. *Risk Traceability Matrix BEFORE Mitigation*

	Impact				
Likelihood	**1**	**2**	**3**	**4**	**5**
5 (Very High)					
4 (High)					
3 (Medium)					
2 (Low)					
1 (Very Low)					

C Additional Technical Data

[Include any supplementary data sheets, extended test results, circuit diagrams, or source-code listings that are too detailed for the main body but are needed for completeness.]